

FutureStrive

Construction Intelligence Platform

Prediction Model Documentation

Crack Intelligence · Defect Volume Intelligence · Reactive Crack Detection

This document provides a complete technical reference for all AI modules in the Construction Intelligence Platform. It covers every input feature, the model used for each prediction target, dataset characteristics, measured accuracy, and the reactive crack detection model wrapper contract.

Version 2.0 · July 2026 · Reactive Detection Release

1. Crack Intelligence Module

Predicts three outputs from 21 concrete pour parameters: (1) crack occurrence probability, (2) severity grade, and (3) crack type. Designed to flag high-risk pours before placing and drive IS 456:2000-aligned corrective action.

1.1 Dataset Overview

Property	Detail
Total Samples	5,000 rows
Feature Count	21 input features
Targets	3 (crack_occurred · crack_severity · crack_type)
Generation	Synthetic — score-based rules derived from IS 456:2000 and IS 7861
Train / Test Split	80% / 20% (4,000 train · 1,000 test)
Random Seed	42

1.2 Input Features

All 21 features are captured at pour level. Categoricals are one-hot encoded; numerics are standardised inside the sklearn pipeline.

Feature	Type	Description
concrete_grade	Categorical	Mix design grade: M25, M30, M35, M40. Higher grade = denser mix = lower permeability and crack risk.
cement_type	Categorical	OPC 43 Grade, OPC 53 Grade, PPC, or PSC. Affects hydration heat and shrinkage behaviour.
admixture_type	Categorical	No Admixture, Naphthalene-based superplasticiser, or Polycarboxylate-based superplasticiser.
water_cement_ratio_design	Numeric	Target W/C from mix design. IS 456:2000 Table 5 specifies exposure-class maxima (e.g., 0.45 for M35 moderate exposure).
water_cement_ratio_actual	Numeric	Measured W/C at site. When actual > design + 0.01 this is the strongest single crack predictor.
wc_ratio_tolerance_spec	Numeric	Maximum allowable W/C in the project quality specification.
target_slump_mm	Numeric	Workability target in mm. Linked to admixture dosage and water content.
max_aggregate_size_mm	Numeric	Nominal maximum aggregate size: 10 mm or 20 mm. Affects water demand.
planned_pour_month	Categorical	Calendar month of the pour. Captures seasonal temperature, humidity, and wind exposure patterns.
curing_method	Categorical	Ponding, Sprinkling, Wet Burlap, or Curing Compound. Affects moisture retention efficiency.
planned_curing_duration_days	Numeric	Specified curing duration in days. IS 456:2000 Section 13.5 requires 14 days minimum when mineral admixtures are used.
actual_curing_duration_days	Numeric	Actual curing achieved at site. Each day below the 14-day minimum increases crack probability.

Feature	Type	Description
spec_min_curing_days	Numeric	Project-specific minimum curing requirement (may exceed IS 456 minimum).
pre_pour_checklist_signed_off_ratio	Numeric	Fraction of pre-pour checklist items signed off (0–1). Low values indicate missed QC hold points.
shrinkage_risk_season	Categorical	Site shrinkage risk classification: LOW, MEDIUM, or HIGH. Derived from site climate data.
wind_exposure_category	Categorical	Sheltered, Normal, or Exposed. High wind accelerates evaporation and plastic shrinkage cracking.
site_environment	Categorical	Inland, Coastal, or Industrial. Coastal / industrial environments increase chloride and carbonation risk.
accessibility	Categorical	Open, Semi-enclosed, or Enclosed. Influences wind speed and temperature at the pour surface.
city	Categorical	Construction city. Maps to climate zone for default temperature and humidity parameters.
project_tier	Categorical	Class A / B / C contractor classification based on past performance audit results.
count_similar_elements	Numeric	Number of structurally similar elements already poured on site. Used to estimate population-level crack recurrence risk.

1.3 Models & Accuracy

Three independent XGBoost pipelines — one per target. All share identical preprocessing: median imputation + StandardScaler (numeric); most-frequent imputation + OneHotEncoder (categorical).

Target	Task	Algorithm & Key Hyperparameters	Test Accuracy
crack_occurred	Binary Classification	XGBClassifier n_estimators=300, max_depth=5 learning_rate=0.05, subsample=0.85	AUC-ROC: 0.94 F1 (weighted): 0.91
crack_severity	Multi-class (4 labels)	XGBClassifier n_estimators=300, max_depth=4 learning_rate=0.05, subsample=0.85	Accuracy: 0.87 F1 (weighted): 0.86
crack_type	Multi-class (5 labels)	XGBClassifier n_estimators=300, max_depth=4 learning_rate=0.05, subsample=0.85	Accuracy: 0.83 F1 (weighted): 0.82

1.4 Prediction Outputs

Output	Classes / Range	Meaning
Crack Probability	0 – 100 %	Probability that a crack will occur. Returned by predict_proba() of the occurrence classifier.
Crack Severity	None · Minor · Moderate · Severe	'None' is shown when probability < 25 %. Driven by curing deficit, W/C excess, and placing temperature.
Crack Type	Plastic Shrinkage · Plastic Settlement · Drying Shrinkage · Thermal · Structural	Dominant crack mechanism predicted from environmental conditions and material parameters.

2. Defect Volume Intelligence Module

Predicts four outputs from 26 project, trade, material, site, and QA/QC features: (1) defects per floor, (2) dominant defect type, (3) severity grade, and (4) root cause. All cost-related fields are explicitly excluded.

2.1 Dataset Overview

Property	Detail
Total Samples	4,000 rows
Feature Count	26 input features
Targets	4 (defect_count · defect_type · severity_grade · root_cause)
Generation	Synthetic — multi-feature score-based rules tied to construction stage, QC compliance, subcontractor class, material status, SPI, and workforce skill ratio
Train / Test Split	80% / 20% (3,200 train · 800 test)
Random Seed	42

2.2 Input Features

Group A — Project & Trade Profile

Feature	Type	Description
project_type	Categorical	Residential, Commercial, Industrial, or Infrastructure. Governs baseline defect rates and inspection frequency.
gfa_sqm	Numeric	Gross Floor Area in square metres. Larger footprint means more concurrent trades and higher defect exposure.
total_floors	Numeric	Total number of floors. Used to contextualise defect count per floor.
structural_system	Categorical	RCC Frame, Shear Wall, Flat Slab, or Steel Frame. Affects which defect type dominates (Structural vs. Finishes).
subcontractor_class	Categorical	Class A / B / C based on historical performance. Class C generates 2–4× more defects per floor than Class A.
past_defect_rate_per_floor	Numeric	Historical average defects per floor from previous projects by the same subcontractor. Strongest predictor of future defect count.
workforce_size	Numeric	Total workers on site. Larger crews increase coordination complexity and supervision demand.
skill_ratio	Numeric	Fraction of workforce classified as skilled (0–1). Below 0.60 significantly increases process-related defects.
site_engineer_experience_yrs	Numeric	Years of experience of the lead site engineer. Below 4 years flags elevated supervision risk.

Group B — Activity & Progress

Feature	Type	Description
construction_stage	Categorical	Foundation, Substructure, Superstructure, MEP Rough-in, Finishes, or Facade. Primary driver of the dominant defect type.

Feature	Type	Description
spi	Numeric	Schedule Performance Index (Earned Value ÷ Planned Value). SPI < 0.90 signals schedule pressure; rushed work directly raises defect rates.
concurrent_activities	Numeric	Number of active trade packages simultaneously on site. Higher values reduce QC attention per activity.
prior_rework_same_element	Binary	1 = element has already been reworked. Prior rework approximately doubles subsequent defect probability on the same element.
qc_hold_point_compliance_pct	Numeric	Fraction of mandatory QC hold points signed off before proceeding (0–1). Below 0.80 is the primary process-defect driver in the model.
third_party_inspection	Binary	1 = client-appointed third-party inspector present. Projects with TPI typically show 30–50% lower defect escape rates.

Group C — Material & Site

Feature	Type	Description
material_grade	Categorical	Concrete or material grade: M25, M30, M35, M40. Tied to mix design compliance requirements.
approved_supplier	Binary	1 = material from a pre-qualified approved supplier. Unapproved suppliers raise material defect risk.
delivery_variance_days	Numeric	Days by which material delivery deviated from programme. Positive values indicate late delivery; negative values indicate early delivery.
test_certificate_status	Categorical	Pass, Fail, or Pending. Using materials with Fail or Pending certificates is a primary material-defect driver.
site_storage_condition	Categorical	Good, Fair, or Poor. Poor conditions degrade material properties before use.
non_productive_days	Numeric	Days lost to weather or other downtime in the reporting period. Values above 7 signal weather-driven defect risk.

Group D — Historical QA/QC

Feature	Type	Description
defects_recorded_to_date	Numeric	Running total of defects raised on this project. A rising count signals systemic process or material failure.
defect_rate_current_project	Numeric	Current project defect rate (defects per floor to date). Compared against portfolio average as a benchmark.
top_defect_type_1	Categorical	Most frequent defect category raised so far: Concrete Defects, Waterproofing, MEP Installation, Tiling/Finishing, or Structural.
defect_closure_rate_pct	Numeric	Fraction of raised defects that have been formally closed (0–1). Low values indicate backlog and systemic QC failure.
portfolio_avg_defect_rate	Numeric	Average defect rate across all active projects in the portfolio. Used as a baseline benchmark for comparison.

2.3 Models & Accuracy

Four independent XGBoost pipelines — one per target. The count pipeline uses XGBRegressor; the three classification pipelines use XGBClassifier with LabelEncoder stored on the pipeline for human-readable inference.

Target	Task	Algorithm & Key Hyperparameters	Test Accuracy
defect_count	Regression	XGBRegressor n_estimators=300, max_depth=5 learning_rate=0.05, subsample=0.85	R ² = 0.918 MAE = 1.33 defects/floor
defect_type	Multi-class (4 labels)	XGBClassifier n_estimators=300, max_depth=4 learning_rate=0.05, subsample=0.85	Accuracy = 99.5% F1 (weighted) = 0.995
severity_grade	Multi-class (4 labels)	XGBClassifier n_estimators=300, max_depth=4 learning_rate=0.05, subsample=0.85	Accuracy = 82.5% F1 (weighted) = 0.826
root_cause	Multi-class (4 labels)	XGBClassifier n_estimators=300, max_depth=4 learning_rate=0.05, subsample=0.85	Accuracy = 88.6% F1 (weighted) = 0.884

2.4 Prediction Outputs

Output	Classes / Range	Meaning
Defect Count	0 – 50 (integer)	Predicted number of defects per floor for the next activity, given current site conditions.
Defect Type	Structural · MEP · Finishes · Facade	Dominant defect category expected. Driven primarily by construction stage and historical top defect type.
Severity Grade	Minor · Moderate · Major · Critical	Minor = cosmetic only. Moderate = rework required. Major = regulatory concern. Critical = stop-work / safety risk.
Root Cause	Process · Material · Supervision · Weather	Primary causal factor. Process = QC failures. Material = supplier or test issues. Supervision = skill or experience gaps. Weather = non-productive days.

3. Shared Pipeline Architecture

Both modules use an identical scikit-learn + XGBoost pattern ensuring reproducible, version-controlled, and auditable predictions.

Stage	Implementation	Purpose
Numeric Imputation	SimpleImputer(strategy='median')	Fills missing numeric values without introducing distribution bias.
Numeric Scaling	StandardScaler	Normalises all numeric features to zero mean and unit variance before tree induction.
Categorical Imputation	SimpleImputer(strategy='most_frequent')	Fills missing categoricals with the modal category observed during training.
Categorical Encoding	OneHotEncoder(handle_unknown='ignore')	One-hot encodes all categoricals. Unknown categories seen at inference are safely ignored.
Column Routing	ColumnTransformer	Routes numeric and categorical features to their respective preprocessing sub-pipelines in parallel.
Model	XGBRegressor or XGBClassifier (tree_method='hist')	Gradient-boosted trees. hist mode enables fast, memory-efficient training on tabular data.
Serialisation	joblib.dump / joblib.load	Preprocessor + model saved as one .joblib file for atomic, zero-drift inference.
Label Encoding	sklearn LabelEncoder stored as pipe.label_encoder_	Attached to each classification pipeline so human-readable labels are recovered at inference with inverse_transform().

4. Saved Model Files

File	Module	Target	Type
crack_occurrence_model.joblib	Crack Intelligence	crack_occurred	XGBClassifier Pipeline
crack_severity_model.joblib	Crack Intelligence	crack_severity	XGBClassifier Pipeline
crack_type_model.joblib	Crack Intelligence	crack_type	XGBClassifier Pipeline
defect_count_model.joblib	Defect Volume Intelligence	defect_count	XGBRegressor Pipeline
defect_type_model.joblib	Defect Volume Intelligence	defect_type	XGBClassifier Pipeline
defect_severity_model.joblib	Defect Volume Intelligence	severity_grade	XGBClassifier Pipeline
defect_rootcause_model.joblib	Defect Volume Intelligence	root_cause	XGBClassifier Pipeline

All .joblib files are excluded from Git via .gitignore. Regenerate locally: `python generate_defect_dataset.py` → `python train_defect_models.py` and the corresponding crack training scripts.

5. Accuracy Summary

Model File	Target	Metric	Score	Rating
crack_occurrence_model	Crack Probability	AUC-ROC	0.940	Excellent
crack_severity_model	Crack Severity	F1 (weighted)	0.860	Very Good
crack_type_model	Crack Type	F1 (weighted)	0.820	Good
defect_count_model	Defects / Floor	R ²	0.918	Excellent
defect_type_model	Defect Type	F1 (weighted)	0.995	Excellent
defect_severity_model	Severity Grade	F1 (weighted)	0.826	Good
defect_rootcause_model	Root Cause	F1 (weighted)	0.884	Very Good

All scores measured on a held-out 20% test split (random_state=42). Excellent ≥ 0.90 | Very Good 0.85–0.89 | Good 0.80–0.84.

6. Reactive Crack Detection Module

The reactive track allows a site photo to be uploaded directly in the AI Assistant. The platform passes the image through `crack_detector.py` — a pluggable model wrapper that enforces a standard output contract so the AI report and chat integration work regardless of the underlying segmentation architecture (PyTorch, ONNX, Keras, etc.).

6.1 Model Output Contract

`run_crack_detection(image)` must always return a dict with these keys. The LLM engine and `app.py` read exclusively from this dict.

Key	Type	Description
crack_detected	bool	True if any crack is found in the image.
confidence	float 0–1	Model confidence score for the top detection.
crack_type	str	Hairline Structural Shrinkage Settlement No Crack
severity_estimate	str	Minor Moderate Severe Critical None
area_fraction	float	% of image pixels classified as crack.
num_instances	int	Number of distinct crack segments detected.
estimated_width_mm	float	Estimated real-world crack width in mm (from model or geometry).
bounding_boxes	list	List of [x1, y1, x2, y2] pixel-coordinate boxes.
annotated_image	PIL.Image	Original image with overlay (bounding boxes / mask). None if no crack.
model_mode	str	'real' for live model, 'stub' for demo mode, 'stub_fallback:...' on error.

6.2 Severity Thresholds (Area Fraction)

Severity	Area Fraction Range	Indicator Colour	Recommended Action
Minor	0 – 1.5%	Green	Monitor — document and re-inspect in 30 days.
Moderate	1.5 – 4.0%	Amber	Repair within 7 days. Apply crack filler or sealant.
Severe	4.0 – 8.0%	Red	Immediate repair. Restrict loading. Notify structural engineer.
Critical	> 8.0%	Purple	STOP WORK. Evacuate area. Structural assessment required.

6.3 IS 456:2000 Permissible Crack Width by Type

Crack Type	IS 456 Width Limit (mm)	Notes
Hairline	0.10	Surface only. No structural implication if ≤ 0.10 mm.
Shrinkage	0.20	Plastic or drying. Exceeding limit requires curing remedy.
Structural	0.20	Flexural or shear. Exceeding limit requires structural review.
Settlement	0.30	Foundation movement. Geotechnical investigation required.

6.4 Connecting the Real Segmentation Model

Three changes are required in `crack_detector.py` when the trained model is ready:

Step	File Location	Change
1	<code>crack_detector.py</code> line 47	Set <code>REAL_MODEL_PATH</code> to the saved model file path (<code>.pt</code> / <code>.h5</code> / <code>.onnx</code>).
2	<code>crack_detector.py</code> line 44	Set <code>USE_STUB = False</code> to disable demo mode.
3	<code>crack_detector.py</code> line 214	Implement <code>_run_real_model(model, image)</code> — run inference and return the standard dict. A PyTorch skeleton is provided in the function docstring.

No changes are required in `app.py`, `intent_router.py`, or `llm_engine.py`. The report, IS code references, and chat integration work automatically from the returned dict.